CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

April 2013

EE203/EE203.01 Electrical Measurement & Measuring Instruments

Date: 25.04.2013, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain in detail different types of phase sequence indicators. [4]

(b) Explain briefly Weston type Synchroscope. [3]

Q - 2 (a) Draw Equivalent circuit and phasor diagram of Current Transformer. Explain why C.T. is not operated with secondary side open circuited. [7]

(b) Explain different methods used for localizing ground faults in underground cables. [7]

OR

Q - 2 (a) Explain briefly difference between C.T. and P.T. State & explain different types of errors in P.T. and methods to reduce these errors. [7]

(b) A C.T. has a bar primary and 200 secondary winding turns. The secondary winding burden is an ammeter of resistance 1.2 Ω and reactance 0.5 Ω, the secondary winding has a resistance of 0.2 Ω and reactance 0.3 Ω respectively. The core requires the equivalent of an mmf of 100 A for magnetization and 50 A for core losses. [7]

(1) Find the primary winding current and ratio error when the ammeter in the secondary winding circuit indicates 5 A.

(2) How many turns could be reduced in the secondary winding in order that ratio error be zero for this condition.

Q - 3 (a) Explain construction of Permanent Magnet Moving Coil type instrument. Also derive it's general torque equation. [7]

(b) Derive general torque equation for Electrodynamometer type instruments. Also explain different types of errors in these instruments. [7]

OR

Q - 3 (a) Explain the operation of Half-wave rectifier type voltmeter. Also explain various factors affecting the performance of rectifier type instruments. [7]

(b) Explain with neat diagram operation of attraction type Moving Iron instrument. Differentiate between PMMC and Moving Iron type instrument. [7]

SECTION – II

Q - 4 (a) State the difference between accuracy and precision of a measurement. [02]

(b) Explain the observational errors in measurement with suitable example. [02]

(c) Three resistors have the following ratings: [03]

$R_1 = 37\ \Omega \pm 5\%$, $R_2 = 75\ \Omega \pm 5\%$, $R_3 = 50\ \Omega \pm 5\%$. Determine the magnitude and limiting error in ohm and in percent of the resistance if these resistances are connected in series.

Q - 5 (a) Explain Kelvin's double bridge method for the measurement of low resistance. [5]

(b) A length of cable is tested for insulation resistance by the loss of charge method. An electrostatic voltmeter of infinite resistance is connected between the cable conductor and earth, forming therewith a joint capacitance of 600 pF. It is observed that after charging the voltage falls from 250 V to 92 V in 1 minute. Calculate the insulation resistance of cable. [2]

(c) What are the different difficulties encountered in the measurement of high voltage resistances? Explain how these difficulties are overcome. [3]

(d) The four arms of a Wheatstone bridge are as follows: [4]

AB=1000 Ω; BC=100 Ω; CD=200 Ω and DA=2005 Ω

The battery has an emf of 5 V and negligible internal resistance. The galvanometer has a current sensitivity of 10 mm/μA and an internal resistance of 100 Ω. Calculate the deflection of galvanometer and the sensitivity of the bridge in terms of deflection per unit change in resistance. Galvanometer is connected across B and D while battery is connected across A and C.

OR

Q - 5 (a) Derive the general equations for balance of an a.c. bridge. [5]

(b) Explain the working principle of Anderson's bridge and also derive its balance equations. [5]

(c) A Maxwell's Inductance comparison bridge with arm AB consists of a coil with induction L_1 and r_1 in series with a non-inductive resistance R. Arm BC and coil CD are each a non-inductive resistance of 100 Ω. Arm AD consists of standard variable inductor L and resistance of 32.7 Ω. Balance is obtained when L_2=47.8mH and R=1.36 Ω. Find the resistance and inductance of the coil in arm AB. Detector is connected across B & D while Supply is connected across A & C respectively. [4]

Q - 6 (a) Explain in detail about the laboratory type DC potentiometer. [7]

(b) Discuss the applications of AC potentiometer. [7]

OR

Q - 6 (a) Describe a method for experimental determination of flux density in a specimen of magnetic material using a ballistic galvanometer. Explain how the correction for flux in the air space between the specimen and the coil is applied. [7]

(b) Explain the method for experimental measurement of magnetizing force acting on a specimen of magnetic material. [7]

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